

Probability HW8

Problem #1

Individual density function: $f(t|\beta) = 2\beta^{-2}te^{-(t/\beta)^2} \quad t > 0, \beta > 0$

Joint density function: $L(\beta) = g(t_1, t_2, \dots, t_n | \beta) = \prod_{i=1}^n f(t_i | \beta) = \prod_{i=1}^n (2\beta^{-2} t_i e^{-(t_i/\beta)^2})$

Log-likelihood: $\ell(\beta) = \log L(\beta) = \sum_{i=1}^n \log(2\beta^{-2} t_i e^{-(t_i/\beta)^2})$
 $= \sum_{i=1}^n [\log 2 - 2 \log \beta + \log t_i - (\frac{t_i}{\beta})^2]$
 $= n \log 2 - 2n \log \beta + \sum_{i=1}^n \log t_i - \sum_{i=1}^n (\frac{t_i}{\beta})^2$

Differentiate log-likelihood: $\frac{d}{d\beta} \ell(\beta) = \frac{d}{d\beta} (-2n \log \beta) - \frac{d}{d\beta} \sum_{i=1}^n (\frac{t_i}{\beta})^2 = -\frac{2n}{\beta} + \frac{2}{\beta^3} \sum_{i=1}^n t_i^2$

$$\text{let } \frac{d}{d\beta} \ell(\beta) = 0 \Rightarrow -\frac{2n}{\beta} + \frac{2}{\beta^3} \sum_{i=1}^n t_i^2 = 0$$

$$\Rightarrow \beta = (\frac{1}{n} \sum_{i=1}^n t_i^2)^{\frac{1}{2}}$$

$$\text{MLE } \hat{\beta} = (\frac{1}{n} \sum_{i=1}^n t_i^2)^{\frac{1}{2}} \quad \text{plug in data } \hat{\beta} \approx \sqrt{\frac{0.25335}{15}} \approx 0.1299 \approx 0.13$$

Problem #2

Individual density function: $f(y_i) = \frac{1}{\sqrt{2\pi}\sigma} \exp[-\frac{(y_i - \mu)^2}{2\sigma^2}] \quad y_i = \log d_i$

Joint density function: $L(\mu, \sigma) = g(y_1, y_2, \dots, y_n | \mu, \sigma) = \prod_{i=1}^n \frac{1}{\sqrt{2\pi}\sigma} \exp[-\frac{(y_i - \mu)^2}{2\sigma^2}]$
 $= (\frac{1}{2\pi})^{\frac{n}{2}} \frac{1}{\sigma^n} \exp[-\frac{1}{2\sigma^2} \sum_{i=1}^n (y_i - \mu)^2]$

log-likelihood: $\ell(\mu, \sigma) = \log L(\mu, \sigma) = -\frac{n}{2} \log(2\pi) - n \log \sigma - \frac{1}{2\sigma^2} \sum_{i=1}^n (y_i - \mu)^2$

Differentiate log-likelihood: $\frac{\partial}{\partial \mu} \ell(\mu, \sigma) = \frac{1}{\sigma^2} \sum_{i=1}^n (y_i - \mu)$

$$\frac{\partial}{\partial \sigma} \ell(\mu, \sigma) = -\frac{n}{\sigma} + \frac{1}{\sigma^3} \sum_{i=1}^n (y_i - \mu)^2$$

$$\text{let } \frac{\partial}{\partial \mu} \ell(\mu, \sigma) = 0 \Rightarrow \hat{\mu} = \frac{1}{n} \sum_{i=1}^n y_i$$

$$\frac{\partial}{\partial \sigma} \ell(\mu, \sigma) = 0 \Rightarrow \hat{\sigma} = \left[\frac{1}{n} \sum_{i=1}^n (y_i - \hat{\mu})^2 \right]^{\frac{1}{2}}$$

$$P(D \geq U) = 0.01 \Rightarrow P(D < U) = 0.99$$

$$P(D < U) = P(\log D < \log U) = P\left(\frac{\log D - \mu}{\sigma} < \frac{\log U - \mu}{\sigma}\right) = \Phi\left(\frac{\log U - \mu}{\sigma}\right) \therefore \Phi\left(\frac{\log U - \mu}{\sigma}\right) = 0.99$$

let $Z_{0.99}$ be the 0.99-quantile of the standard normal distribution

$$\frac{\log U - \mu}{\sigma} = Z_{0.99} \Rightarrow U = \exp(\mu + Z_{0.99} \sigma) \therefore \hat{U}_{100} = \exp(\hat{\mu} + Z_{0.99} \hat{\sigma})$$

$$y_i = \log d_i \quad i = 1, 2, \dots, 37$$

$$\hat{\mu} = \frac{1}{37} \sum_{i=1}^{37} y_i \approx 8.561$$

$$\hat{\sigma} = \left[\frac{1}{37} \sum_{i=1}^{37} (y_i - \hat{\mu})^2 \right]^{\frac{1}{2}} \approx 0.4725$$

$$\begin{aligned} \hat{U}_{100} &= \exp(\hat{\mu} + Z_{0.99} \hat{\sigma}) = \exp(8.561 + 2.33 \times 0.4725) \\ &= \exp(9.662) \\ &\approx 15751 \text{ ft}^3/\text{s} \end{aligned}$$

Problem #3

$$X_i = \mu + \varepsilon_i \quad \varepsilon_i \sim N(0, \sigma^2) \quad \sigma = 0.1 \text{ mg}$$

$$\therefore X_i \sim N(\mu, \sigma^2) \quad \sigma = 0.1 \text{ mg}$$

$$\bar{X} = \frac{1}{5} \sum_{i=1}^5 X_i = \frac{1}{5} (3.142 + 3.163 + 3.155 + 3.150 + 3.141) = 3.1502 \text{ mg}$$

$$\text{The } (1-\alpha) \text{ CI is } [\bar{X} - Z_{1-\frac{\alpha}{2}} \frac{\sigma}{\sqrt{n}}, \bar{X} + Z_{1-\frac{\alpha}{2}} \frac{\sigma}{\sqrt{n}}]$$

(a) 95% CI

$$Z_{0.975} \frac{\sigma}{\sqrt{n}} = 1.96 \times \frac{0.1}{\sqrt{5}} \approx 0.0877$$

$$\therefore 95\% \text{ CI} : 3.1502 \pm 0.0877 = [3.0625, 3.2379]$$

(b) 99% CI

$$Z_{0.995} \frac{\sigma}{\sqrt{n}} = 2.5758 \times \frac{0.1}{\sqrt{5}} \approx 0.1152$$

$$\therefore 99\% \text{ CI} : 3.1502 \pm 0.1152 = [3.0350, 3.2654]$$

Problem #4

For large samples, by CLT, $\bar{X} \hat{\sim} N(\mu, \frac{\sigma^2}{n}) = N(\mu, \frac{11.3^2}{81})$

$$90\% \text{ CI} : [\bar{X} - Z_{0.95} \frac{\sigma}{\sqrt{n}}, \bar{X} + Z_{0.95} \frac{\sigma}{\sqrt{n}}]$$

$$\bar{X} = 74.6 \quad Z_{0.95} \frac{\sigma}{\sqrt{n}} = 1.645 \times \frac{11.3}{\sqrt{81}} \approx 2.07$$

$$90\% \text{ CI} : 74.6 \pm 2.07 = [72.53, 76.67]$$

Problem #5

For large samples, by CLT, $\bar{X} \hat{\sim} N(\mu, \frac{\sigma^2}{n}) \quad \sigma = 400$

$$95\% \text{ CI} : \bar{X} \pm Z_{0.975} \frac{\sigma}{\sqrt{n}}$$

$$\text{The margin of error: } E = Z_{0.975} \frac{\sigma}{\sqrt{n}} \leq 60$$

$$1.96 \frac{400}{\sqrt{n}} \leq 60$$

$$n \geq 170.8$$

$\therefore$ sample size must be integer

$\therefore n = 171$ (round up)